DM Week 1
Anatomy PLG

A 62-year-old man presents to the emergency department with sudden onset, severe epigastric pain radiating to the mid-back. The pain began after a heavy meal and alcohol intake and is associated with nausea and vomiting. Physical examination reveals epigastric tenderness without rebound. Serum lipase is markedly elevated. CT imaging shows an inflamed pancreas with peripancreatic fat stranding.

Which nerve pathway most directly transmits visceral pain from the affected organ to the spinal cord?

A. Pelvic splanchnic nerves (S2–S4)
B. Greater thoracic splanchnic nerves (T5–T9)
C. Lesser thoracic splanchnic nerves (T10–T11)
D. Lumbar splanchnic nerves (L1–L2)
E. Pudendal nerve

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The pancreas is a foregut associated organ therefore the visceral pain fibers are carried with the sympathetic fibers which are the greater thoracic splanchnics

A 59-year-old woman presents with progressive abdominal distension, constipation, and intermittent crampy abdominal pain. Imaging reveals a volvulus involving the distal sigmoid colon. She describes the pain as deep, dull, and poorly localized.

Which spinal cord segments are most likely receiving visceral afferent pain signals from the affected portion of bowel?

A. T5–T9
B. T10–T11
C. L1–L2
D. S2–S4
E. L4–S1

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The distal sigmoid is associated with the hindgut and therefore transmits it pain fibers through the lumbar splanchnic nerves

A 67-year-old man with long-standing atherosclerosis presents with acute onset lower abdominal pain and bloody diarrhea. Colonoscopy reveals ischemia of the distal sigmoid colon below the pelvic pain line.

Where would the patient most likely localize his pain?

A. Periumbilical region
B. Epigastrium
C. Sacral and perineal region
D. Right upper quadrant
E. Left shoulder

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C. Sacral and perineal region
D. Right upper quadrant
E. Left shoulder

The distal sigmoid is associated with the hindgut and therefore below the pelvic pain line transmits it pain fibers through the pelvic splanchnic nerves (S2-S4) which out of the options will correlate most with the sacral and perineal regions

A 48-year-old man undergoes surgery for rectal cancer. Postoperatively, he has loss of voluntary bowel control but intact reflex relaxation of the internal anal sphincter.

Damage to which nerve most likely explains this deficit?

A. Pelvic splanchnic nerve

B. Hypogastric nerve

C. Pudendal nerve

D. Lumbar splanchnic nerve

E. Inferior hypogastric plexus

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Voluntary bowel control is achieved by the external anal sphincter (skeletal muscle) which is innervated by the inferior rectal nerve which is a branch of the pudendal nerve

A 35-year-old woman with chronic constipation reports difficulty initiating defecation. Rectal exam reveals normal voluntary sphincter contraction but failure of internal sphincter relaxation.

Which autonomic pathway normally mediates relaxation of the internal anal sphincter?

A. Sympathetic fibers from lumbar splanchnic nerves
B. Parasympathetic fibers from pelvic splanchnic nerves
C. Somatic fibers from pudendal nerve
D. Vagal parasympathetic fibers
E. Enteric neurons exclusively

A 35-year-old woman with chronic constipation reports difficulty initiating defecation. Rectal exam reveals normal voluntary sphincter contraction but failure of internal sphincter relaxation.

Which autonomic pathway normally mediates relaxation of the internal anal sphincter?

A. Sympathetic fibers from lumbar splanchnic nerves

B. Parasympathetic fibers from pelvic splanchnic nerves

C. Somatic fibers from pudendal nerve

D. Vagal parasympathetic fibers

E. Enteric neurons exclusively

The internal anal sphincter is innervated by the pelvic splanchnic nerves

A 54-year-old man with alcoholic cirrhosis presents with progressive abdominal distension and weight gain. On exam, there is a fluid wave and shifting dullness. Paracentesis confirms ascites.

In which anatomic space is this fluid accumulating?

A. Between parietal peritoneum and abdominal wall fascia
B. Between visceral peritoneum and abdominal organs
C. Between parietal and visceral peritoneum
D. Within the retroperitoneum
E. Within mesenteric fat

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D. Within the retroperitoneum
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The liver is an intraperitoneal organ therefore fluid from the liver will leak into the peritoneal cavity which is defined as the space between the visceral and parietal peritoneum

A 70-year-old man presents with severe postprandial abdominal pain and weight loss. Angiography shows occlusion of the superior mesenteric artery.

Which portion of the gastrointestinal tract is most affected?

A. Distal esophagus
B. Proximal duodenum
C. Jejunum
D. Descending colon
E. Upper rectum

A 70-year-old man presents with severe postprandial abdominal pain and weight loss. Angiography shows occlusion of the superior mesenteric artery.

Which portion of the gastrointestinal tract is most affected?

A. Distal esophagus - foregut
B. Proximal duodenum - foregut
C. **Jejunum - midgut (SMA supplies midgut)**
D. Descending colon - hindgut
E. Upper rectum - hindgut

A 55-year-old man develops massive upper GI bleeding from a posterior duodenal ulcer.

Which artery has most likely been eroded?

A. Proper hepatic artery
B. Cystic artery
C. Gastroduodenal artery
D. Left gastric artery
E. Superior mesenteric artery

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Most duodenal ulcers are in the first part of the duodenum due to the high acid content from the stomach. The posterior wall of the proximal duodenum sits right against structures in the retroperitoneal wall. A rupture will erode into vessels behind it. The gastroduodenal artery runs posterior to the first part of the duodenum so if an ulcer on the posterior duodenal wall keeps digging, the first major hit is the gastroduodenal artery.

During laparoscopic cholecystectomy, a surgeon isolates the cystic artery within Calot’s triangle.

Which structure forms the superior boundary of this triangle?

A. Common bile duct
B. Cystic duct
C. Inferior vena cava
D. Liver edge
E. Portal vein

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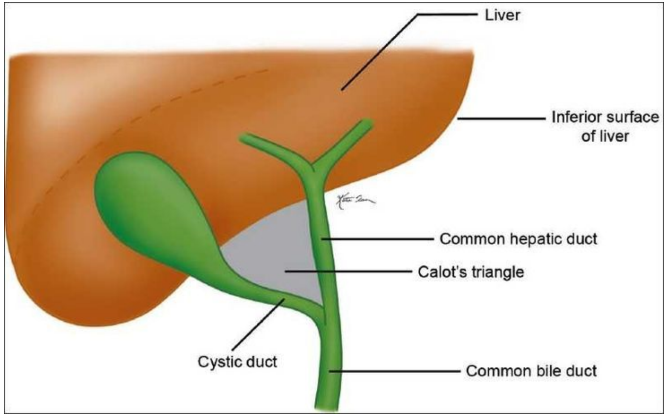

A surgeon performing exploratory laparotomy identifies the lesser omentum connecting the stomach to the liver.

Which ligaments form the lesser omentum?

A. Gastrocolic and gastrosplenic

B. Hepatogastric and hepatoduodenal

C. Gastrosplenic and gastrophrenic

D. Gastrocolic and gastrophrenic

E. Falciform and round ligament

A surgeon performing exploratory laparotomy identifies the lesser omentum connecting the stomach to the liver.

Which ligaments form the lesser omentum?

A. Gastrocolic and gastrosplenic - greater omentum

B. Hepatogastric and hepatoduodenal

C. Gastrosplenic and gastrophrenic - greater omentum

D. Gastrocolic and gastrophrenic - greater omentum

E. Falciform and round ligament - in liver

A 24-year-old woman with rapid weight loss develops postprandial vomiting. Imaging shows compression of the third part of the duodenum between the aorta and superior mesenteric artery. Which anatomic change predisposes to this condition?

A. Elongation of ligament of Treitz
B. Shortened ligament of Treitz
C. Increased retroperitoneal fat
D. Inferior displacement of SMA
E. Enlarged pancreas

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Shortened ligament of treitz will pull the duodenum upward and posteriorly. Tethers it more tightly between the aorta and SMA. Makes compression more likely and more severe

A 6-week-old human embryo is examined as part of a research study on early gastrointestinal development. During this stage, the primitive gut tube has formed as the embryo undergoes cephalocaudal and lateral folding. The epithelial lining of the esophagus, stomach, small intestine, and large intestine is differentiating from one of the three primary germ layers. Disruption of this developmental process is known to cause congenital malformations affecting nutrient absorption and mucosal integrity throughout most of the gastrointestinal tract. Which embryological structure gives rise to the epithelial lining of most of the gastrointestinal tract?

A. Ectoderm

B. Neural crest cells

C. Endoderm of the umbilical vesicle

D. Splanchnic mesoderm

E. Intermediate mesoderm

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A. Ectoderm = epithelium of cranial and caudal ends of the gut tube

B. Neural crest cells = neural component

C. Endoderm of the umbilical vesicle = most of GI tract epithelium

D. Splanchnic mesoderm = smooth muscle and connective tissue (gland stroma)

E. Intermediate mesoderm = urogenital system

During normal embryologic development, the intestinal lumen is temporarily occluded due to rapid epithelial proliferation and later restored through a process known as recanalization. Failure of recanalization of the intestinal lumen most directly results in which of the following?

A. Volvulus

B. Bowel atresia

C. Bowel stenosis

D. Omphalocele

E. Hirschsprung disease

During normal embryologic development, the intestinal lumen is temporarily occluded due to rapid epithelial proliferation and later restored through a process known as recanalization. Failure of recanalization of the intestinal lumen most directly results in which of the following?

A. Volvulus = abnormal intestinal rotation

B. Bowel atresia = absence of lumen from complete failure of recanalization

C. Bowel stenosis = narrowed lumen from partial recanalization

D. Omphalocele = failure of midgut return to the abdominal cavity

E. Hirschsprung disease = failed neural crest cell migration

During embryological development, a 7-week-old fetus is forming the primitive gut tube, which is suspended within the peritoneal cavity by mesenteries. The ventral mesentery is present only in the foregut region, and anchors the developing stomach and proximal duodenum to the ventral body wall. As development proceeds, this ventral mesentery differentiates into several adult peritoneal ligaments that help support the liver and stomach. The ventral mesentery of the foregut ultimately becomes which of the following structures?

A) Falciform lig, coronary ligs, and lesser omentum

B) Lesser and greater omentum

C) Falciform ligament and greater omentum

D) Greater omentum and transverse mesocolon

E) Coronary ligs and splenorenal ligament

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A) Falciform lig, coronary ligs, and lesser omentum

B) Lesser and greater omentum = greater omentum arises from dorsal mesentery

C) Falciform ligament and greater omentum

D) Greater omentum and transverse mesocolon = both from dorsal mesentery

E) Coronary ligs and splenorenal ligament = splenorenal from dorsal

A full-term newborn develops bilious vomiting within hours of birth and is unable to tolerate feeds. The pregnancy was notable for polyhydramnios on prenatal ultrasound. Physical exam shows mild epigastric distention without palpable masses. An abdominal radiograph reveals two large air-filled structures in the upper abdomen with absence of distal bowel gas. The condition is caused by a congenital obstruction related to abnormal embryologic development of the gut lumen. Which of the following is the most likely diagnosis?

A. Duodenal stenosis

B. Annular pancreas

C. Duodenal atresia

D. Pyloric stenosis

E. Midgut volvulus

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What is this sign?

A. Duodenal stenosis = partial recanalization, less acute with some distal gas

B. Annular pancreas = duodenal obstruction, some distal gas, not the classic sign

C. Duodenal atresia

D. Pyloric stenosis = presents at 2–6 weeks with nonbilious projectile vomiting

E. Midgut volvulus = bilious vomiting w corkscrew or malrotation on radiograph

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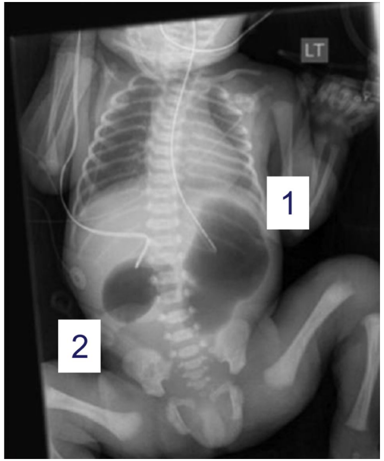

Double bubble

4-week-old male infant is brought to clinic for recurrent vomiting after feeds. The vomiting is forceful and projectile, occurs shortly after feeding, and is non-bilious. On abdominal examination, a small, firm, olive-shaped mass is palpated in the right upper quadrant. Which of the following acid-base abnormalities is most likely present?

A. Metabolic acidosis

B. Respiratory acidosis

C. Metabolic alkalosis

D. Respiratory alkalosis

E. Mixed acidosis

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A. Metabolic acidosis

B. Respiratory acidosis

C. Metabolic alkalosis = persistent vomiting leads to loss of gastric HCl

D. Respiratory alkalosis

E. Mixed acidosis

As the intestines rotate and return to the abdominal cavity, certain portions of the gastrointestinal tract become fixed to the posterior abdominal wall. These organs are described as secondarily retroperitoneal, meaning they were initially intraperitoneal but later fused to the posterior body wall during development. Which structure becomes secondarily retroperitoneal at the end of gut rotation?

A. Jejunum

B. Ileum

C. Transverse colon

D. Duodenum

E. Sigmoid colon

As the intestines rotate and return to the abdominal cavity, certain portions of the gastrointestinal tract become fixed to the posterior abdominal wall. These organs are described as secondarily retroperitoneal, meaning they were initially intraperitoneal but later fused to the posterior body wall during development. Which structure becomes secondarily retroperitoneal at the end of gut rotation?

A. Jejunum = intraperitoneal

B. Ileum = intraperitoneal

C. Transverse colon = intraperitoneal

D. Duodenum = secondarily retroperitoneal

E. Sigmoid colon = intraperitoneal

A 2-day-old male newborn is evaluated for intermittent abdominal distention and feeding intolerance. Review of embryological development highlights the normal connection between the midgut and the yolk sac via the omphaloenteric (vitelline) duct, which typically involutes by the 7th week of gestation. Persistence of the proximal omphaloenteric duct results in which of the following structures?

A. Omphaloenteric cyst

B. Umbilical hernia

C. Urachal fistula

D. Meckel diverticulum

E. Omphaloenteric fistula

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A. Omphaloenteric cyst = cyst forms in fibrous cord of remnant duct

B. Umbilical hernia = incomplete closure of the umbilical ring

C. Urachal fistula = failure of obliteration of the urachus (allantois)

D. Meckel diverticulum

E. Omphaloenteric fistula = full duct patent (fecal discharge at umbilicus)

A newborn presents with bilious vomiting shortly after birth. Imaging reveals partial obstruction of the second portion of the duodenum and an annular pancreas. This embryological abnormality is caused by which of the following processes?

A. Failure of dorsal pancreatic bud formation

B. Incomplete fusion of pancreatic ducts

C. Bifid ventral pancreatic bud encircling the duodenum

D. Failure of duodenal recanalization

E. Abnormal stomach rotation

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A newborn fails to pass meconium within 48 hours and has abdominal distension. Contrast enema shows a transition zone between a dilated proximal colon and a narrowed distal segment. The most likely embryological defect involves which of the following?

A. Endodermal proliferation

B. Gut rotation

C. Neural crest cell migration

D. Lateral folding

E. Recanalization

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B. Gut rotation

C. Neural crest cell migration

D. Lateral folding

E. Recanalization

What is the name of this condition?

Hirschsprung disease

During midgut development, the gut loop rotates how many degrees total around the superior mesenteric artery?

A. 90° clockwise

B. 180° clockwise

C. 270° counterclockwise

D. 360° clockwise

E. 180° counterclockwise

During midgut development, the gut loop rotates how many degrees total around the superior mesenteric artery?

A. 90° clockwise

How does the stomach move?

B. 180° clockwise

C. 270° counterclockwise = 3 x 90 degree rotations

D. 360° clockwise

E. 180° counterclockwise

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A. 90° clockwise

B. 180° clockwise

C. 270° counterclockwise

D. 360° clockwise

E. 180° counterclockwise

How does the stomach move?

1. 90 clockwise on longitudinal (head to toe)
   (left surface faces the front)
2. Tilt on short axis (front to back)
   (proximal goes left and down)
   (distal goes right and up)